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An automated analytics approach for diabetic retinopathy detection with ensemble deep learning models in healthcare

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ABSTRACT

Diabetic Retinopathy (DR) is a leading complication of prolonged diabetes, which poses a significant threat to vision and may lead to permanent blindness. Early identification and timely intervention are crucial to preventing disease progression. Traditionally, DR diagnosis relies on medical examination of retinal fundus images by expert ophthalmologists, which is time-consuming and resource intensive. However, deep learning techniques, particularly medical imaging, have demonstrated remarkable performance in the automated detection and classification of DR. This study proposes an ensemble-based deep learning framework using feature-level fusion stacking, which integrates four complementary convolutional neural networks named ReXInDen and three complementary convolutional neural networks named ReXDen for automated DR detection from retinal fundus images. These frameworks extract high-level features from each backbone, concatenate them into a unified representation, and classify using a feedforward neural network. Three datasets were utilized to validate the model including a region-specific dataset collected from Bangladeshi medical sources. The proposed ReXInDen model achieved accuracies of 98.27%, and 98.69% on Dataset 1 and Dataset 2, while ReXDen achieved the highest accuracy of 99.05% on Dataset 3. These results indicate a substantial improvement over individual models and demonstrate the potential of the ensemble approach to support early-stage DR detection. Moreover, these models show promise for integration into automated DR screening tools that can aid in reducing the global burden of diabetic vision loss.

1. Introduction

Diabetic Retinopathy (DR) is a debilitating ailment that mostly affects individuals with diabetes, especially those who experience extended or uncontrolled blood sugar episodes [1]. It is a major global source of vision impairment and blindness raising serious concerns about public health. It can be triggered by chronic hyperglycemia in several ways. Long-term high blood glucose levels harm the retinal blood vessels by a confluence of inflammatory, vascular, and metabolic processes. The microvascular difficulties that define diabetic retinopathy are ultimately caused by this damage, which also raises oxidative stress, impairs blood flow regulation, and breaks down the blood-retinal barrier [2,3]. Gradually, the condition weakens the retina, which absorbs visual data and sends it to the brain. Its rigidity can show up in a variety of ways, from minor anomalies to serious consequences like complete loss of sight and retinal detachment.

Diabetes has grown to be one of the diseases with the fastest rate of expansion in recent decades. Around the world, 2.6% of blindness cases are attributable to diabetic retinopathy [4]. Retinal imaging diagnosis of DR is based on a sophisticated set of precise characteristics and locations. According to the World Health Organization (WHO), 422 million patients are living with diabetes, and the prevalence of the disease has increased from 4.7% to 8.0% worldwide as noted in [5]. There are estimates that 191 million individuals globally will have diabetes by 2030 [6,7]. The retinal images that reveal the blood vessels, retina, optic disc, and macula allow for the identification of early signs of DR, such as microaneurysms, hemorrhages, and retinal exudates. Fig. 1 shows the effects of diabetic retinopathy on retinal structure and highlights the key changes.

The detection and timely treatment of DR are required in preventing irreversible vision loss [8]. Every year a large number of people in the world are diagnosed with diabetes, especially in developed countries like Bangladesh [9], where efficient approaches for DR screening is crucial. However, the current methods of identification of DR are mainly by ophthalmoscopic examinations. These manual methods are time-consuming, require expert ophthalmologists, and are often inaccessible in resource-limited settings [10].

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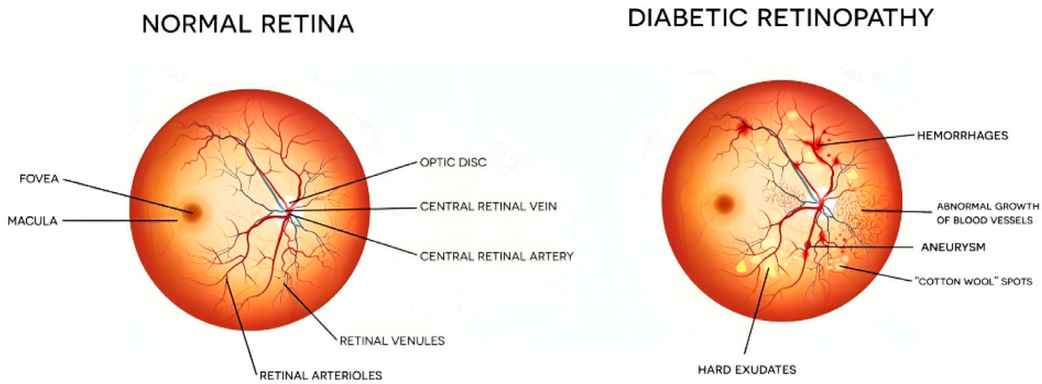

Fig. 1. Comparative Retinal Fundus Images Illustrating the Effects of Diabetic Retinopathy (DR). Left: Normal retinal fundus. Right: Retinal fundus showing pathological features associated with DR [16].

Implementing the automation of retinal image-based diabetes diagnosis in a clinical setup can significantly enhance the detection rate of DR and eliminate the challenges posed by traditional methods. Recent advancements of artificial intelligence (AI), particularly in deep learning (DL) have demonstrated promising potential in automating medical image analysis tasks including retinal disease detection. Several DL methods including Convolutional Neural Networks (CNNs) have shown outstanding performance in image classification, segmentation, and object detection, making them suitable for DR diagnosis using retinal fundus images. Additionally, multiple studies have developed CNN-based models for DR detection. Despite the success of various deep learning approaches, many existing models face limitations such as overfitting, class imbalance, and limited generalization across datasets. Moreover, most studies rely on a single deep learning architecture, which may fail to capture the complex and heterogeneous patterns of diabetic retinopathy in fundus images. To address these challenges, ensemble learning has emerged as a powerful strategy that combines the strengths of multiple models to improve classification accuracy, robustness, and reliability.

S. Paivitra et al. [1] use deep learning models on fundus images to improve classification accuracy through the use of ensemble learning. VGG19, ResNet50, and InceptionV3 models are used to extract the features. Although effective, it faces high computational cost, risk of overfitting, and limited real-world generalizability due to dependence on a public dataset. Inamullah et al. [11] employ an ensemble model that includes three Convolutional Neural Networks. Augmentation enhances data diversity, but the ensemble's use of varied datasets adds complexity and processing load. Over-diversification may cause overfitting, and its generalizability beyond the APTOS-2019 dataset remains unproven. F.M.J. Mehedi Shamrat et al. [12] developed DRNet13, an accurate and efficient model using advanced preprocessing like median filtering and gamma correction, along with eight data augmentation techniques.

However, this study proposed an ensemble-based deep learning framework for automated DR detection, introducing two variants: ReXInDen, which integrates ResNet18, Xception, InceptionV3, and DenseNet201, and ReXDen, which integrates ResNet18, Xception, and DenseNet201. These models were utilized for effective feature extraction from the input images, and the extracted features were concatenated into a single comprehensive feature vector. These ensemble approaches follow a stacking via feature-level fusion strategy, where the unified feature representation is then fed into a Feedforward Neural Network (FNN) that serves as the final DR classifier. This design leverages the complementary strengths of the CNN backbones while enabling the FNN to perform robust classification based on the enriched fused features. Several data preparation techniques such as image pre-processing and data augmentation, were applied to enhance the performance of the proposed model. Additionally, enhancement methods such as Contrast Limited Adaptive Histogram Equalization (CLAHE), kernel filtering, bilateral filtering, and Laplacian filtering were applied to increase the contrast and sharpness of the structures of retinal images. Furthermore, we address existing gaps in current DR detection approaches by assessing performance on diverse datasets, including a unique dataset of retinal images from the Bangladeshi population.

Experiments were conducted on three datasets to evaluate the effectiveness of the proposed ensemble-based deep learning framework for DR detection. Dataset 1 is a publicly available retinal fundus image dataset that is widely used for DR research [13]. Dataset 2 was collected by our team and collaborators from medical institutions in Bangladesh. Dataset 3 is a combined dataset that combines dataset 1 and dataset 2 to create a more diverse dataset. On Dataset 1, the ReXInDen framework achieved a classification accuracy of 98.27%. For Dataset 2, the ReXInDen model yielded a slightly higher accuracy of 98.74%, showing strong performance on region-specific data. At last, the ReXDen ensemble approach achieved the best performance on Dataset 3, with an accuracy of 99.05%. These findings show the potential of the proposed ensemble approach in developing tools for DR, which can support automated retinal screening systems.

The main contributions of our research can be summarized as follows.

- We proposed an ensemble-based deep learning framework for binary classification of DR from retinal fundus images. These models integrate the CNN architectures such as ResNet18, InceptionV3, Xception, and DenseNet201 for effective feature extraction, followed by a Feedforward Neural Network for final classification.
- One of the key contributions of this work is the use of a unique dataset obtained from two hospitals in Bangladesh. By combining this dataset with a publicly available one, the proposed approach improves the model's accuracy in diverse populations and imaging conditions.
- Through comprehensive experiments and comparative analysis with other methods, the proposed ensemble model shows remarkable performance in binary DR classification, achieving a highest accuracy of 99.05% on the combined dataset.

This paper is organized as follows: Provides a comprehensive Literature review in Section 2. Outlines the methodology in Section 3. Offers the experimental results of this study in Section 4. Finally, Section 5 encapsulates the conclusion and recommendations for future works.

2. Related works

The past few years have witnessed significant interest in the development of automated tools for DR diagnosis. Artificial intelligence (AI) is used to analyze fundus photos to detect symptoms of DR with several benchmark methods. AI lends great accuracy to the DR diagnosis systems. Numerous studies have evaluated the accuracy of AI-driven DR diagnosis systems in real-world scenarios, which is promising.

2.1. CNN-based single model approaches

The goal behind B. Ihnaini et al. [14] study was to build a highly accurate model for classifying fundus images for diagnostic purposes in diabetes. The research includes Xception and EfficientNetB3 models using APTOS and IDRiD datasets. To improve classification performance, these models were used as a single model after they were integrated into an ensemble. The integrated model achieved 97.47% accuracy, which is the highest in this study. The study's findings have some limitations in scope, as the model was trained and tested basically on the APTOS and IDRiD datasets, which might be affected by its generalizability to other patient groups or imaging circumstances.

Devendra et al. [15] used transfer learning with VGG16 and VGG19 as feature extractors, feeding those features into classical classifiers; Logistic Regression performed best. Their top result was 90.4% accuracy using VGG16 with Logistic Regression. This study used a small, single-source dataset where only 282 images are available, and reliance on VGG feature-based classical classifiers without comparison to end-to-end state-of-the-art models.

María Herrero et al. [16] proposed a CNN-based deep learning framework by using ResNet50 for diabetic retinopathy classification. The model was trained and validated on APTOS 2019, Messidor, Messidor-2, EyePACS, and DDR datasets. It achieved the highest accuracy of 97.83% in binary classification. The study relied only on public datasets without external clinical validation, had limited testing on diverse populations, and achieved lower accuracy in binary classification.

Raiaan et al. [17] proposed a lightweight CNN-based architecture, RetNet-10, for the detection of diabetic retinopathy, aiming to balance accuracy with computational efficiency. The model was trained on a combined dataset and achieved a classification accuracy of 98.14%, outperforming several transfer learning models such as VGG16, ResNet50, and InceptionV3. Despite its strong performance, the study highlighted some limitations, including limited dataset diversity, risk of overfitting due to augmentation strategies, and the need for validation on larger and more heterogeneous datasets to ensure broader generalization

To enhance the classification of diseases in retinal imagery, this study [18] addressed some issues related to the DenseNet121, Xception, ResNet50, VGG16 and VGG19 CNN structures. According to the researchers, their CNN model is more accurate and easier to understand than other competing methods. Both binary and multi-class classification results were very good, with an accuracy of 95.27% that surpassed the 82% achieved by Xception in binary classification. The data was often imbalanced and images had different qualities, making the results of predictions less reliable.

Roberto et al. [19] introduced a deep learning-based approach for diabetic retinopathy detection using a custom CNN architecture trained on the Kaggle EyePACS dataset. Their model achieved an overall 95.68% accuracy, with strong performance in binary classification. The study highlighted key limitations such as dataset imbalance, reliance on a single dataset (EyePACS), and reduced generalization to other publicly available datasets.

2.2. Ensemble based approaches

G. T. Reddy et al. [20] proposed an ensemble machine learning model to improve the accuracy of diabetic retinopathy classification. In their study, the model leveraged the combined strengths of multiple classifiers such as Random Forest, Decision Tree, AdaBoost, K Nearest Neighbor, and Logistic Regression to improve detection performance on a dataset of diabetic retinopathy. The ensemble approach achieved an accuracy of 82%, sensitivity of 85% and specificity of 79%. Although the dataset is relatively small, the results presented here are somewhat limited in scope, especially regarding the model's robustness and generalizability.

Sashank et al. [21] proposed a binary classification model with an ensemble approach using AdaBoost classifiers to identify the presence of DR. To enable more detailed DR stage prediction, they combined an ensemble of ResNet models and an artificial neural network as a meta-classifier; thus, they can use the ANN for multiclass classification of DR stages. The model was tested on binary classification and had an accuracy of 78.9%, a precision of 86%, a recall of 79% and an F1 score of 79%, which is promising. In addition, the researchers pointed out that advanced pre-processing as well as using larger datasets for further improvements.

Kasim et al. [22] proposed a diabetic retinopathy early detection method using MobileNet transfer learning combined with ensemble learning. The approach was validated on Messidor, APTOS 2019, and DDR datasets, with class imbalance addressed using the Synthetic Minority Oversampling Technique (SMOTE). The method achieved early detection accuracies of 98.6% (Messidor), 88.95% (APTOS 2019), and 97.46% (DDR). The study focused mainly on early-stage detection rather than comprehensive DR grading, relied on synthetic data generation via SMOTE, which may introduce artifacts, and was only validated on three public datasets without testing generalizability on real-world clinical data.

S. Mondal et al. [23] laid out an automated ensemble model to increase detection accuracy with two deep learning models, namely modified DenseNet101 and ResNeXt, which are ensembled for the detection of diabetic retinopathy. Data augmentation using MixGAN is used to rectify the imbalance in class representation. The model exceeded current models, reaching 96.98% accuracy with precision and recall of 97% for binary classification. The model's abstractions include class imbalance, a small dataset size, and challenges with binary classification accuracy.

A. Desiani et al. [24] used ensemble learning to overcome the limitations of the ResNet-50, MobileNet, and EfficientNet CNN models with the goal of enhancing the classification of the disease from retinal images. They sought to reduce overfitting and improve accuracy by merging these models. With an F1-score of 93.42%, a Cohen's kappa score of 0.866, 93.3% accuracy, 93.51% sensitivity, and 93.32% specificity, their ensemble model technique produced noteworthy measures of performance. They observed some challenges, for example, the high expenses associated with ensemble learning models.

Peddapullaiahgari et al. [25] proposed DiaRetULS-Net, an ensemble model for the detection of diabetic retinopathy. In this study integrated CLAHE preprocessing, Discrete Wavelet Transform, and Local Binary Patterns for feature extraction, with a hybrid architecture combining U-Net for segmentation. The model evaluated on the Messidor-2 dataset and achieved 98.83% accuracy, 98.87% specificity, and 99.21% sensitivity

Tăbăcaru et al. [26] aimed at classifying diabetic retinopathy using ensemble learning. Gamma correction for contrast manipulation was done using Image processing. Feature extraction by using Shannon and fuzzy entropies was also used for the data. This study achieved 92.9% in classification accuracy, 90.2% in F1 score, and 94.1% in AUC in DR classification. They comparatively produce lower performance using the proposed model.

2.3. Limitations of existing works and research gap

Despite the rapid advancement of deep learning methods for diabetic retinopathy detection, existing works continue to face several limitations that restrict their clinical applicability. Many models have been trained on small-scale or imbalanced datasets such as APTOS and EyePACS, which limits their robustness when applied to diverse populations. Overfitting remains a critical challenge, especially in models that rely heavily on augmentation or synthetic data generation, as these approaches may not capture real-world variations in retinal images. Furthermore, a large portion of prior studies utilized single CNN architectures, which often fail to capture the complex and heterogeneous manifestations of DR, leading to weaker generalization. Ensemble methods have been attempted, but most rely on homogeneous architectures or simple voting strategies. Many of them still struggle with computational overhead, limited external validation, and inadequate testing on region-specific data, which collectively raise concerns about their scalability and real-world adoption.

To address these shortcomings, our work introduces an ensemble learning framework that integrates CNN architectures that are not only powerful individually but also complementary in the features they extract. In this study, we selected four diverse models, ResNet18, InceptionV3, Xception, and DenseNet201 for each chosen for its unique strengths in capturing retinal patterns. ResNet18 contributes residual learning for hierarchical features, DenseNet201 promotes dense connectivity for feature reuse, Xception leverages depthwise separable convolutions for efficient yet complex feature extraction, and InceptionV3 captures multi-scale contextual information through its multi-branch structure. Instead of relying on voting, this research employs feature-level concatenation, which merges the rich and distinct feature maps from each model into a unified representation. This strategy creates a highly discriminative feature space that better reflects both micro-level lesions and macro-level retinal structures. By validating the framework on both a public dataset and a region-specific Bangladeshi dataset. Advanced preprocessing and image enhancement techniques were also applied to improve retinal clarity and mitigate the impact of noisy or low-quality images. Through this design, our models not only alleviate overfitting and dataset bias but also demonstrate superior robustness, achieving 99.05% accuracy on the combined dataset. Thus, the proposed approach directly addresses the gaps in prior research and demonstrates improved robustness, generalizability, and clinical relevance in automated DR detection.

3. Materials and methods

Detecting diabetic retinopathy is crucial for preventing vision loss among people with diabetes. DR can lead to anopia if left untreated, and traditional diagnosis through manual retinal image examination is time-intensive and prone to human error, especially in large-scale screening. Advances in deep learning and computer vision provide an opportunity to develop automated, accurate tools to address this challenge [27].

This study focuses on building an ensemble-based deep learning approach for the automated detection of diabetic retinopathy from retinal fundus images. The experimental workflow comprises several components such as data pre-processing, model development using deep learning architectures, and evaluation using standard performance metrics. Fig. 2 presents a detailed workflow diagram illustrating the overall methodology, highlighting each stage from data pre-processing to model evaluation. Three datasets were utilized for training and testing the models. To build up the ensemble model, the CNN-based deep learning models such as ResNet18, InceptionV3, Xception, and DenseNet201 were fine-tuned and integrated using an ensemble strategy to enhance the robustness of classifications. Each model in this ensemble has its own strengths in the detection of DR. ResNet18 contains residual connections that facilitate the model to learn deeper, more meaningful features. Based on a multi-branch structure, InceptionV3 realizes fine as well as broad image details. With depthwise separable convolution, Xception is able to capture complex patterns very effectively with a low computational cost and DenseNet201's dense connections propagate the information throughout the layers so it can learn subtle variation of retinal images. All of these models combine to make a comprehensive ensemble that outperforms what any single model could achieve in picking out the signs of DR. The following subsections describe all steps involves in this research.

Table 1
Summary of the three datasets showing total images, number of DR images, and number of No DR images.

	Dataset 1	Dataset 2	Dataset 3
No of patients	No data	1280	Combination
No of Images	2838	2630	5468
DR images	1408	910	2318
NO-DR images	1430	1720	3150

3.1. Dataset description

3.1.1. Dataset 1: Kaggle diabetic retinopathy detection dataset (KDRD)

To develop and evaluate the proposed ensemble-based diabetic retinopathy (DR) detection framework, three datasets were utilized. The first dataset was obtained from the publicly available Kaggle DR Detection dataset [13]. This dataset contains 2838 high-resolution retinal fundus images labeled in two classes: DR and No DR as this study focuses on binary classification. A balanced method was used by grouping the data into a training set, validation set and test set, where each set contained images with and without DR. The definition of all the variables in the dataset is detailed in Table 1 below.

3.1.2. Dataset 2: Bangladeshi Diabetic Retinopathy Imaging Dataset (BD-DRiD)

This research used a second dataset from Bangladesh that was specifically prepared for this research. The dataset consists of 2630 retinal fundus images, each discussed in Table 1. This dataset was independently collected by our research team in collaboration with two medical institutions in Bangladesh: Barishal Diabetics Hospital and Mymensing Medical College. Expert ophthalmologists with 8–10 years of clinical experience annotated the images into two categories: DR and No DR. This dataset enables the model to learn from region-specific clinical presentations, potentially improving diagnostic accuracy and generalizability for the Bangladeshi population.

3.1.3. Dataset 3: Combined dataset of KDRD and BD-DRiD

The third dataset was constructed by merging Dataset 1 and Dataset 2 to create a more diverse and balanced training corpus. Prior to merging, Random Oversampling class balancing techniques were applied to the training set to mitigate data imbalance and ensure equitable representation across DR severity grades. This combined dataset was used to evaluate whether increased data diversity and volume could enhance the performance of the DR detection model. This holistic approach allows the model to flexibly infer data at many scales, leading to a strong performance at identifying DR in a Bangladeshi context. The enhanced dataset supports stronger model performance and greater potential utility in local healthcare settings. Table 1 presents the details of Dataset 3.

Although Random Oversampling was applied to address class imbalance during dataset preparation, a comparative analysis of model performance with and without this technique was not conducted due to computational and time constraints. Specifically, training multiple deep CNN architectures and their hybrid configurations on high-resolution retinal images required substantial GPU resources and extended training time, and repeating the full experimental pipeline for ablation analysis would have significantly increased computational cost beyond the scope of this study. Nevertheless, prior studies have shown that such balancing methods typically enhance model generalization by preventing bias toward the majority class. Future work will include an ablation analysis to quantitatively evaluate its specific impact.

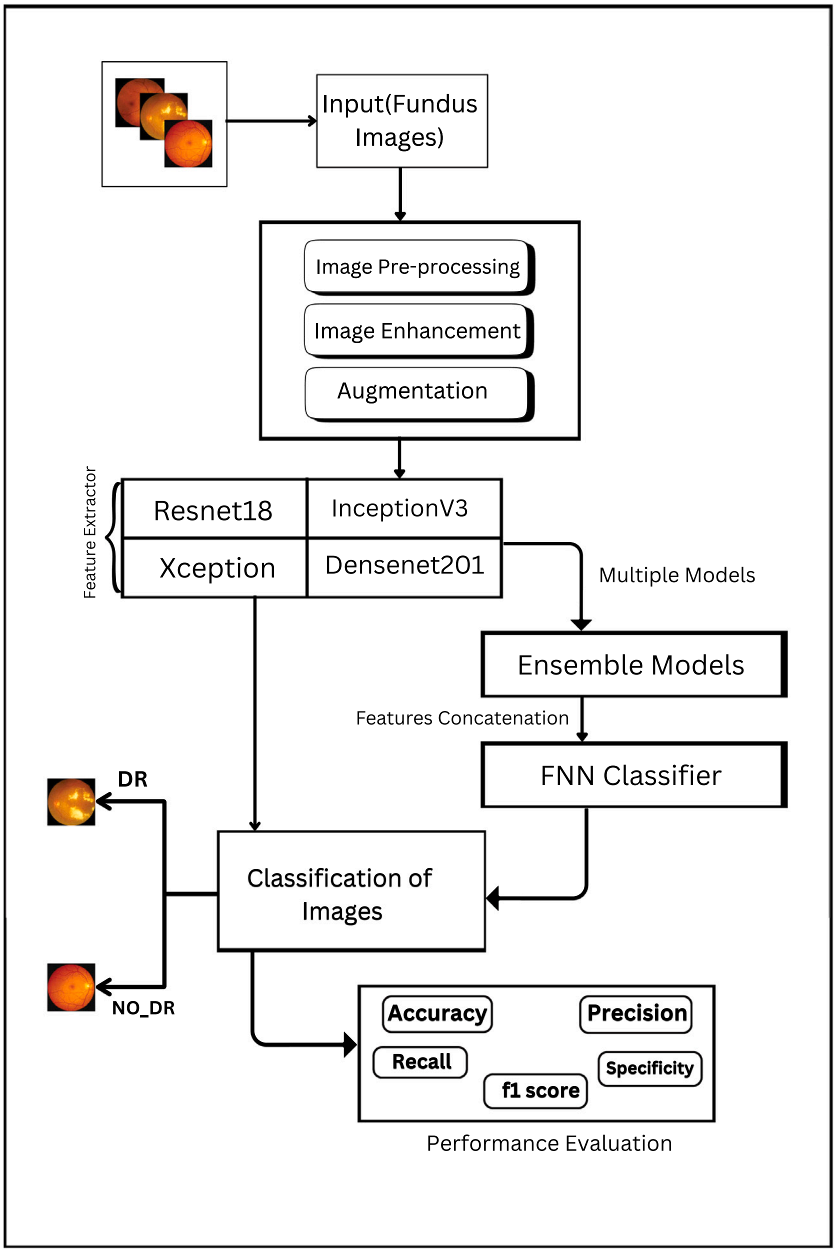

Fig. 2. Overview of the proposed ensemble-based deep learning workflow for DR detection, encompassing data input, data pre-processing, feature extraction and individual model training, ensemble construction, and performance evaluation.

3.2. Image pre-processing

3.2.1. Image resizing

Consistency is important and so images are routinely being resized to 224 × 224 pixels making them the proper size when fed into CNNs such as ResNet which expect consistent input sizes. Eq. (1) shows the image is resized to this specific size, with the transforms function. But the other issue occurs with the original image not being square, then this operation can slightly distort the aspect ratio because it does directly resize the image rather than cropping or padding. It is important to resize as CNNs are very sensitive to input size variations, and without proper resizing, we will get errors and inconsistent results on training or evaluation ?. The Inception model needs to be resized to 299 × 299 pixels.

$$I_{\text{resized}} = \text{Resize}(I_{\text{original}}, (TargetWide, TargetHight)) \tag{1}$$

3.2.2. Tensor conversion

The transforms function converts images from formats like NumPy arrays or PIL images into PyTorch tensors, the essential data structure for deep learning models in PyTorch [28]. This transformation scales the image pixel values from their original 0 to 255 range (for RGB channels) to a normalized range of 0 to 1, making the data more suitable for neural network processing. The transformation occurs mathematically and is shown in Eq. (2). This conversion is crucial because neural networks process data in batches of tensors, and this step ensures the images are ready for efficient model training and inference.

$$X_{\text{tensor}} = \frac{\text{ToTensor}(X)}{255} \rightarrow (C, H, W) \tag{2}$$

3.2.3. Normalization

This study used the transforms Normalize function to adjust the range and distribution of pixel values to a standardized scale, ensuring that each RGB channel has zero mean and unit variance [29]. For each

Table 2
Representation of the datasets after applying the Augmentation technique.

	Dataset 1	Dataset 2	Dataset 3
No of patients	No data	1280	Combination
No of Images	2838	3211	6049
DR images	1408	1104	2512
NO-DR images	1430	2107	3537

channel, the transformation subtracts a predefined mean and divides by the standard deviation shown in Eq. (3). This process improves the consistency of the input data, ensuring that the network sees images with a range similar to what it was trained on. Normalization also speeds up training and enhances stability by preventing any one feature from dominating the learning process due to its larger scale.

$$\text{Normalized value} = \frac{\text{Original value} - \text{Mean}}{\text{Standard Deviation}} \tag{3}$$

3.2.4. Dataset loading and data-loader preparation

The ImageFolder class in PyTorch is designed for loading image data where the folder names represent class labels. The DataLoader wraps the data after it is loaded for fast handling. It does batch processing, speeds up training time and reduces memory resource usage. The DataLoader also shuffles the data to prevent memorizing the order of samples and does parallel processing on the data loading speed. All of these features facilitate convenient data handling, accelerated training and superior generalization of the model.

Raw photos are unsuitable for deep learning models because they differ in size, aspect ratio, and format without preprocessing. Errors may result from dimensions that do not match the desired format and pixel values that range from 0 to 255.

3.3. Augmentation

Augmentation increases dataset diversity and improves model generalization. Techniques such as random horizontal flip, random vertical flip, and random rotation are particularly important for improving the ability of machine learning models. After augmentation, the DR images of the BD-DRiD dataset are increased by almost 200 and the No DR images are increased by almost 400 shown in the table Table 2.

3.4. Image enhancement

For diabetic retinopathy image enhancement, the objective is to increase the contrast and sharpness of the retinal structures visible in images for diagnostics and further analysis. Thus, the Contrast Limited Adaptive Histogram Equalization (CLAHE) algorithms are used in order to enhance the local contrast of the image to make microaneurysms or hemorrhages on the retina more distinguishable [30]. Noise reduction is a critical step in documenting DR images. Important details like vascular structures and lesions are preserved when the image is smoothed by filters Bilateral Filtering or Median Filtering. These improvements make it easier to extract the most important features to aid in the early diagnosis of diabetic retinopathy to ensure that the disease is diagnosed at the right stage for management [31]. In Fig. 3 shows the retinal image before and after enhancement with the Histogram representation along with RGB color channels.

1. **Contrast Limited Adaptive Histogram Equalization (CLAHE)**

$$H'(i) = \min\left(H(i), \text{clip\_limit}\right) \tag{4}$$

$$I'(x,y) = \frac{\text{CDF}(I(x,y))}{\max(\text{CDF})} \cdot (L-1) \tag{5}$$

Where: $H(i)$: Intensity histogram of the local region. clip_limit: Maximum allowable histogram value to limit contrast. $\text{CDF}(I(x,y))$: Cumulative distribution function for intensity $I(x,y)$. L: Number of intensity levels (e.g., 256 for 8-bit images).

2. **Bilateral Filter (Edge-preserving smoothing)**

$$I'(x,y) = \frac{\sum_{(x',y')} G_s(\|p-p'\|) \cdot G_r(|I(p)-I(p')|) \cdot I(p')}{\sum_{(x',y')} G_s(\|p-p'\|) \cdot G_r(|I(p)-I(p')|)} \tag{6}$$

Where: $G_s(d) = \exp\left(-\frac{d^2}{2\sigma_s^2}\right)$: Spatial Gaussian kernel, based on pixel distance. $G_r(r) = \exp\left(-\frac{r^2}{2\sigma_r^2}\right)$: Range Gaussian kernel, based on intensity difference. p, p': Pixel positions. σ_s: Controls spatial smoothing. σ_r: Controls intensity smoothing.

3. **Laplacian Filter for Final Sharpening**

$$L(x,y) = \frac{\partial^2 I}{\partial x^2} + \frac{\partial^2 I}{\partial y^2} \tag{7}$$

$$I'(x,y) = I(x,y) + \beta \cdot L(x,y) \tag{8}$$

Where: $L(x,y)$: Laplacian of the image, representing second-order derivatives.
β: Sharpening factor to control the intensity of edges.

4. **Histogram Calculation**

$$H_k(i) = \sum_{x,y} \delta(I_k(x,y) - i) \tag{9}$$

Where: $H_k(i)$: Frequency of intensity i in channel k ($k \in \{R, G, B\}$ for color images).
$\delta(x)$: Dirac delta function:

$$\delta(x) = \begin{cases} 1, & \text{if } x = 0 \\ 0, & \text{otherwise} \end{cases}$$

$I_k(x,y)$: Intensity of channel k at position (x,y).

3.5. Single model architecture and feature extraction

For diabetic retinopathy detection applications, ResNet, Xception, Inception, and DenseNet are notable models due to their ability to capture subtle pathological patterns in retinal images. Each of these architectures has unique design strengths.

In this study, these architectures are utilized to analyze retinal fundus images, and their complementary characteristics are later exploited through an ensemble framework to improve detection reliability [32]. The architectures and applications of the four CNN models used in this study are described below.

3.5.1. ResNet18

The architecture of ResNet models tackles the vanishing gradient issue with residual learning and depends on residual blocks which make the network learn differences instead of simply mapping the input. The blocks provide stability and better performance to deep networks because they add shortcut or skip connections, which ensure that gradients are correctly calculated during training [33]. To include hierarchical features, the network has four sets of residual blocks with increasing filter sizes, as shown in Fig. 4. In every residual block, there are two convolutional layers, batch normalization and ReLU activations. The network ends with a fully connected layer and global average pooling, which are intended for 1000-class ImageNet classification [34].

3.5.2. InceptionV3

A highly effective and precise model that is frequently used for image classification tasks is InceptionV3, which is a member of Google's Inception family of CNNs. Building on earlier iterations (InceptionV1 and V2), it integrates important improvements including auxiliary classifiers, factorized convolutions, and architectural changes to increase accuracy and efficiency while lowering computational costs [35].

The main component of InceptionV3 is the Inception module, which

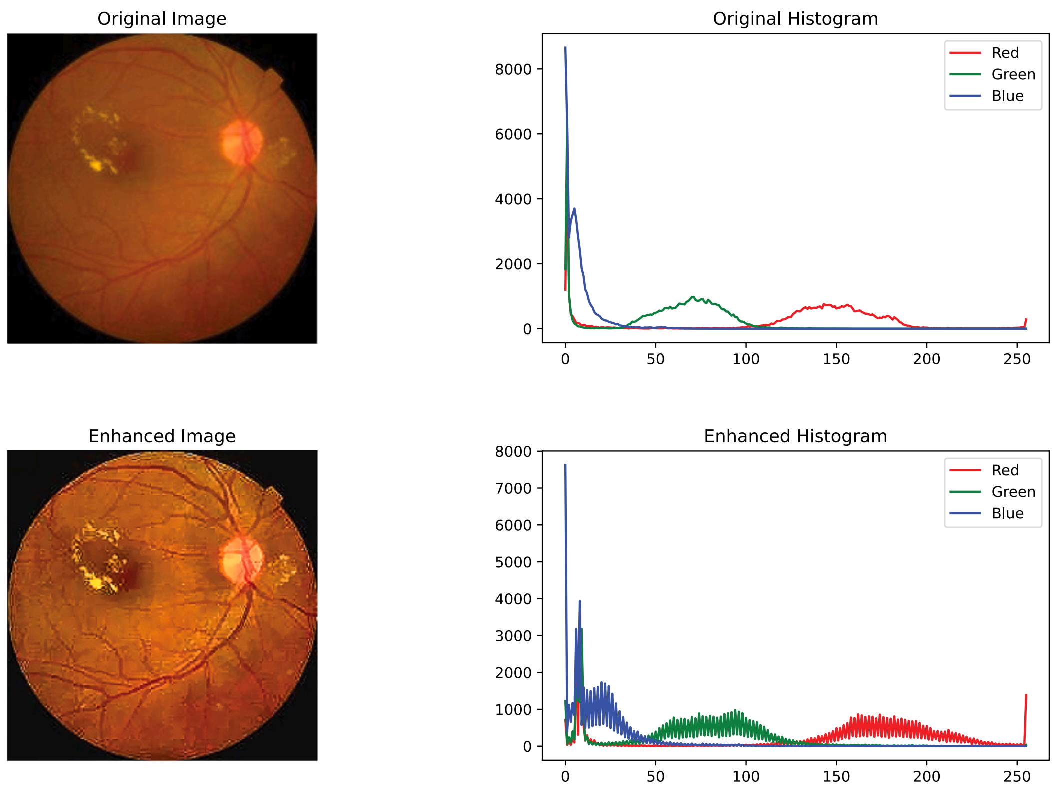

Fig. 3. Visual and histogram representations of retinal images before and after applying the image enhancement technique. The upper row shows the original images, and the lower row displays the enhanced results.

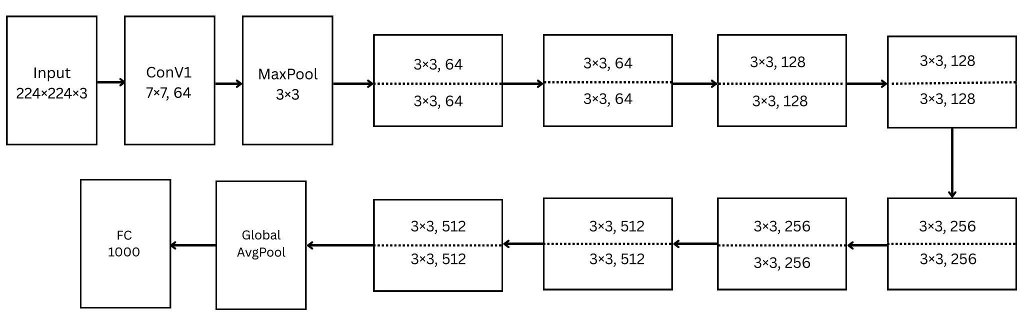

Fig. 4. Architecture of ResNet18 model.

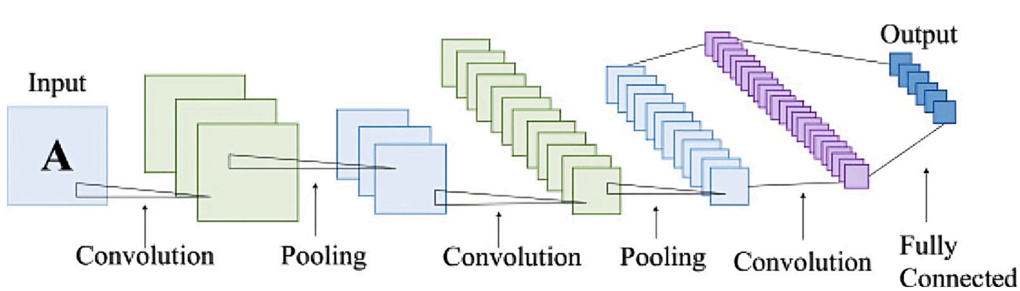

Fig. 5. Architecture of InceptionV3 model.

concatenates the outputs of several filters applied simultaneously to the same input. To effectively extract hierarchical characteristics, inception modules employ a range of filter sizes and branches [36]. By adding intermediate supervision along with improving gradient flow and training stability, auxiliary classifiers offer regularization. Fig. 5 shows a global average pooling layer, a fully connected layer, and softmax activation for classification round out the model.

3.5.3. *Xception*

In 2017, François Chollet introduced the Xception model, a highly sophisticated convolutional neural network that expands on the Inception architecture by substituting depthwise separable convolutions for conventional convolutions. This change is known as "Extreme Inception" since it drastically lowers the amount of parameters and calculations while preserving excellent accuracy.

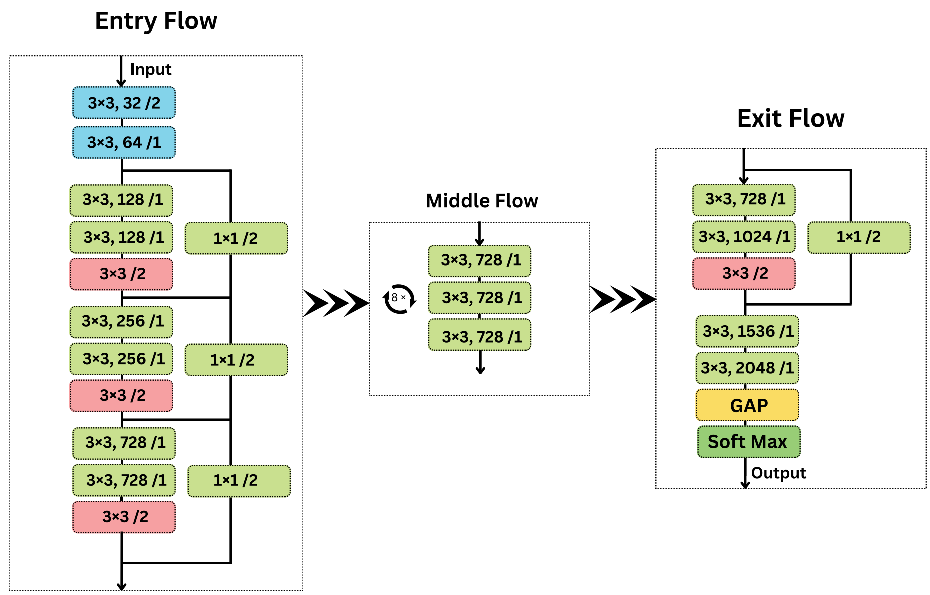

Fig. 6. Architecture of Xception model.

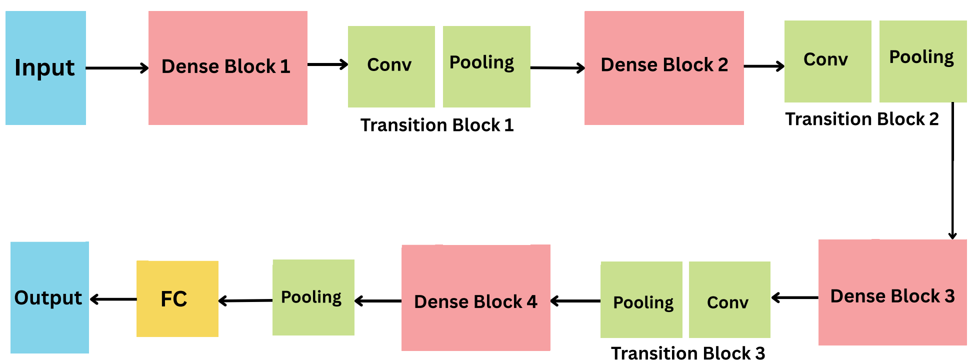

Fig. 7. Architecture of Densenet201 model.

Fig. 6 makes it clear that the Xception model consists of Entry Flow, Middle Flow and Exit Flow. The first convolutions are designed to pick up low-level data in the Entry Flow. The Middle Flow seeks to identify complex features while conserving the image's shape. The Exit Flow process is completed by using depthwise separable convolutions, global average pooling, and a final fully connected softmax layer for classification [37]. Applying residual connections for enhanced stability and gradient flow is crucial, as is using depthwise separable convolutions for quicker and simpler computation.

3.5.4. DenseNet201

Dense connections are the basis of DenseNet201 which is a deep convolutional neural network that was introduced in 2017. Every layer is affected by data above it and passes its data to layers below it. The approach fixes the vanishing gradient challenge and speeds up learning because it makes sure that important features are reused and the gradient flows smoothly.

The network starts from a convolutional layer and then progresses to max pooling and batch normalization, shown in Fig. 7. Transition layers, which reduce spatial dimensions and control the number of channels, divide the four dense blocks that comprise DenseNet201 [38]. When strong feature retention and efficient sharing of information are needed, as with diabetic retinopathy identification, this design is very useful since fine visual details from retinal pictures play a key role in correct analysis. DenseNet201 is suited for working with complex picture processing in deep learning because it easily extracts image features [39,40].

3.6. Ensemble model and feature extraction

To enhance the robustness and accuracy of diabetic retinopathy detection, an ensemble learning framework was developed by integrating multiple convolutional neural network backbones. Each backbone network was initialized with ImageNet pre-trained weights and subsequently fine-tuned on the retinal fundus datasets in order to capture discriminative and domain-specific features. The individual CNNs independently generated high-level feature representations, which were then concatenated into a single comprehensive feature vector as shown in Fig. 8.

The concatenated features were forwarded to a Feedforward Neural Network classifier, which served as the final decision-making component of the proposed framework. The architecture of the FNN shown in Fig. 9 was designed with three fully connected layers. The first hidden layer consisted of 512 neurons activated by the Rectified Linear Unit (ReLU) function, followed by a Dropout layer with a probability of 0.5 to reduce the risk of overfitting. The second hidden layer comprised 256 neurons with ReLU activation, again followed by a Dropout layer ($p = 0.5$). Finally, the output layer contained two neurons, corresponding to the two target classes (DR and No_DR), activated by the Softmax function to produce probabilistic predictions.

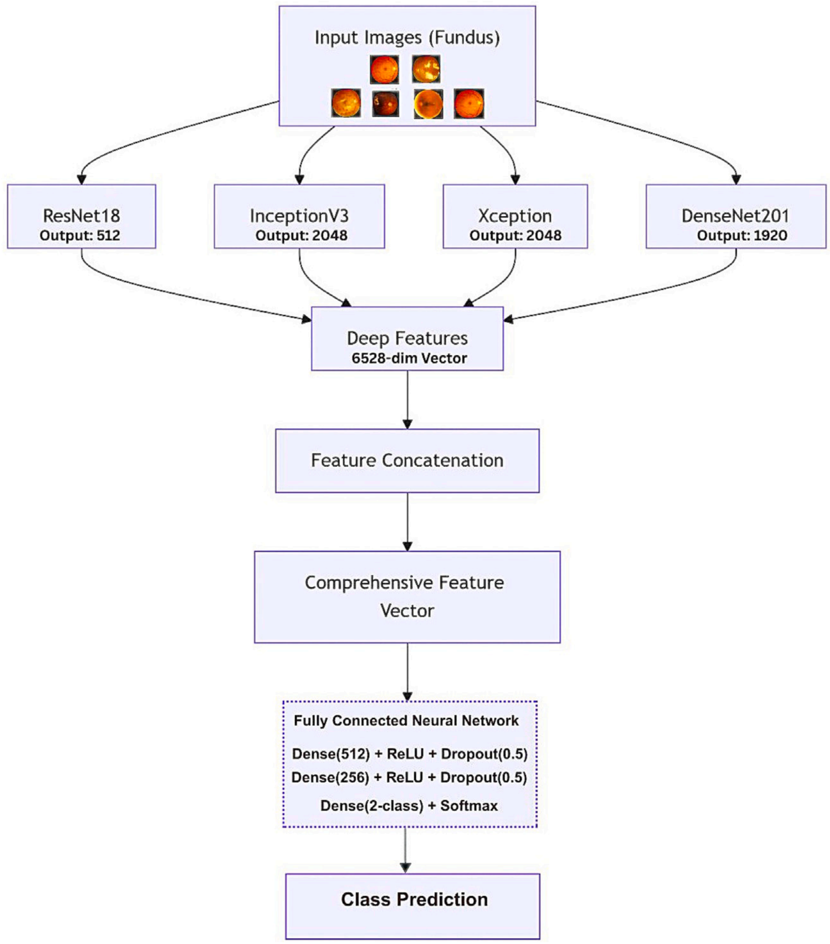

Fig. 8. Architecture of the Proposed Ensemble Model.

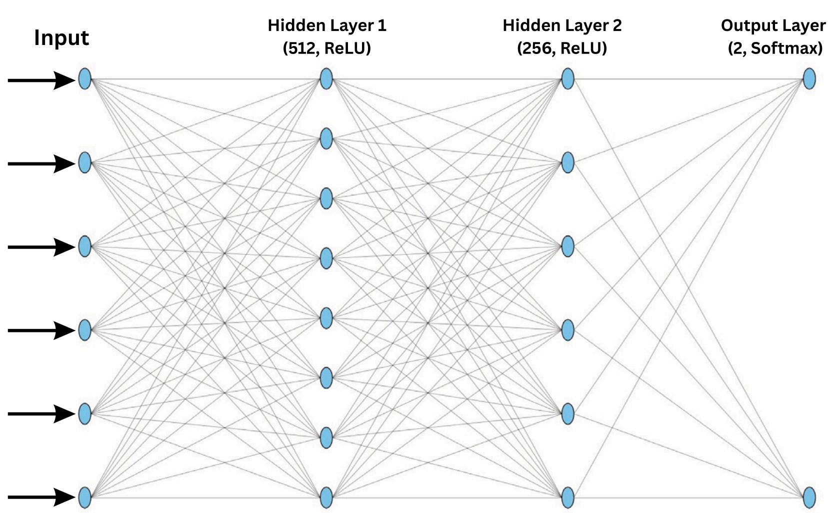

Fig. 9. Architecture of FNN Classifier.

3.6.1. *ReXDen model*

The ReXDen model is a three-network ensemble that integrates ResNet18, Xception, and DenseNet201. The selection of these backbones is based on their complementary capabilities in extracting discriminative retinal features. ResNet18 provides residual hierarchical learning that stabilizes training and captures layered features. Xception leverages depthwise separable convolutions for efficient yet complex pattern extraction, allowing the detection of subtle micro-lesions. DenseNet201 contributes dense connectivity and feature reuse, ensuring gradient flow and capturing fine-grained retinal vessel details. By combining these three models, ReXDen achieves a balanced trade-off between computational efficiency and feature diversity.

Algorithmic Workflow

Input:

Test dataset $D = (X_i, y_i)_{i=1}^{N}$, where X_i is the input image and y_i is the ground truth.

Initialization:

1. Load pre-trained models: ResNet18, Xception, and DenseNet201.
2. Modify the final classification layers for binary DR and No-DR classification.
3. Preprocess images: resize to 224 × 224 pixels, normalize, and convert into tensors.
4. Set testing parameters: batch size, device (CPU/GPU), and data loader for test batches.

Feature Extraction: Let F_{Res18} represent the feature extraction for ResNet18, F_{Xcep} represent the feature extraction function for Xception, and $F_{Dense201}$ represent the feature extraction for DenseNet201.

For an input image X_i, the extracted features from ResNet18 can be denoted as

$$X_i^{Res18} = F_{Res18}(X_i) \tag{10}$$

Similarly, for Xception,

$$X_i^{Xcep} = F_{Xcep}(X_i) \tag{11}$$

And again for DenseNet201,

$$X_i^{Dense201} = F_{Dense201}(X_i) \tag{12}$$

In order to get a single feature vector, concatenate the features that were taken from all four models:

$$X_i^{ReXDen} = \text{concat}(X_i^{Res18}, X_i^{Xcep}, X_i^{Dense201}) \tag{13}$$

Depending on the merged feature vector, a feedforward neural network classifier C is used to predict the class label, where W represents the classifier's weights and b represents the bias term. The classifier's output may be computed as follows:

$$Output_i = C(X_i^{ReXDen}) = softmax(W \cdot X_i^{ReXDen} + b) \tag{14}$$

Finally, the predicted class is determined by applying the $\arg\max$ function over the output logits:

$$Prediction = \arg\max(Output_i) \tag{15}$$

3.6.2. ReXInDen model

The ReXInDen model extends the ReXDen design by including InceptionV3, creating a four-backbone ensemble composed of ResNet18, Xception, InceptionV3, and DenseNet201. This combination was chosen to incorporate multi-scale contextual learning. InceptionV3 captures both global and local retinal features through its multi-branch structure, making it well-suited to identifying lesions of varying sizes and shapes. Together with ResNet18's residual learning, Xception's efficient fine-grained extraction, and DenseNet201's feature reuse, ReXInDen achieves stronger generalizability across datasets and imaging conditions.

Algorithmic Workflow

Input:

Test dataset: $D = \{(X_i, y_i)\}_{i=1}^{N}$, where X_i is the input image, and y_i is the ground truth label.

Pre-trained models: ResNet18, InceptionV3, DenseNet201, and Xception.

Initialization:

1. Load the pre-trained models (ResNet18, InceptionV3, DenseNet201, and Xception).
2. Adjust the final layers of each model to handle binary classification.
3. Load the test dataset with image transformations: 224 × 224 size for ResNet18, DenseNet201, and Xception. 299 × 299 size for InceptionV3.
4. Set batch size, device (CPU/GPU), and data loader.

Feature Extraction: Let F_{Res18} represent the feature extraction for ResNet18, $F_{IncepV3}$ represent the feature extraction function for InceptionV3, F_{Xcep} represent the feature extraction function for Xception, and $F_{Dense201}$ represent the feature extraction for DenseNet201.

For an input image X_i, the extracted features from ResNet18 can be denoted as

$$X_i^{Res18} = F_{Res18}(X_i) \tag{16}$$

For Xception,

$$X_i^{Xcep} = F_{Xcep}(X_i) \tag{17}$$

Similarly, for InceptionV3,

$$X_i^{IncepV3} = F_{IncepV3}(X_i) \tag{18}$$

And again for DenseNet201,

$$X_i^{Dense201} = F_{Dense201}(X_i) \tag{19}$$

In order to get a single feature vector, concatenate the features that were taken from all four models:

$$X_i^{ReXInDen} = \text{concat}(X_i^{Res18}, X_i^{Xcep}, X_i^{IncepV3}, X_i^{Dense201}) \tag{20}$$

Depending on the merged feature vector, a feedforward neural network classifier C is used to predict the class label, where W represents the classifier's weights and b represents the bias term. The classifier's output may be computed as follows:

$$Output_i = C(X_i^{ReXInDen}) = softmax(W \cdot X_i^{ReXInDen} + b) \tag{21}$$

Finally, the predicted class is determined by applying the $\arg\max$ function over the output logits:

$$Prediction = \arg\max(Output_i) \tag{22}$$

The two ensemble configurations, ReXDen and ReXInDen, were designed to systematically evaluate the trade-offs between model diversity and computational efficiency. ReXDen integrates three complementary backbones to balance accuracy with reduced computational cost, making it suitable for real-time or resource-limited screening environments. In contrast, ReXInDen extends this framework by incorporating InceptionV3, which introduces multi-scale contextual feature learning and further enriches the fused representation space. While ReXInDen tends to achieve marginally higher diagnostic accuracy due to the additional feature diversity, ReXDen demonstrates greater efficiency and stability across runs. By pre-specifying and evaluating both ensembles under identical experimental protocols, this study shows that feature-level fusion of complementary CNN architectures consistently enhances diabetic retinopathy detection, and the choice between ReXDen and ReXInDen can be guided by application-specific requirements, whether prioritizing speed and efficiency or maximizing classification accuracy.

3.7. Training and implementation

The dataset was divided into three subsets with a ratio of 73% for training, 17% for validation, and 10% for testing. This split ensured sufficient data for model learning while maintaining a fair evaluation protocol.

All experiments were implemented using the PyTorch framework on a workstation equipped with an AMD Ryzen-7 3800 CPU, 64 GB RAM, and an AMD Radeon RX 580 GPU. Each CNN backbone (ResNet18, DenseNet201, InceptionV3, and Xception) was initialized with ImageNet pre-trained weights. The final classification layers were replaced with binary output layers for DR versus No-DR detection, and all convolutional layers were fine-tuned during training. For the ensemble framework, the feature vectors extracted from each backbone were concatenated and passed through a Feedforward Neural Network (FNN), which produced the final classification. The Adam optimizer was employed with an initial learning rate of 0.001. The ReduceLROnPlateau scheduler was used to automatically reduce the learning rate by a factor of 0.1 when the validation loss did not improve for five consecutive epochs. This dynamic adjustment stabilized training and helped avoid convergence to poor local minima. The CrossEntropyLoss function was applied as the objective function, with a mini-batch size of 64. Training was conducted for up to 30 epochs, and early stopping was introduced to prevent overfitting when validation performance plateaued.

During training, both loss and accuracy were continuously monitored on the training and validation sets to ensure convergence and guide scheduler adjustments. The performance of the trained models was then evaluated on the held-out test set. Random seeds were fixed in NumPy and PyTorch to enhance reproducibility, and the results reported represent the average of multiple runs to reduce variability. Model performance was assessed using various evaluation metrics, including accuracy, precision, recall, F1-score, specificity, and AUC. Confusion matrices and ROC curves were also generated to provide additional insight into classification behavior.

In summary, the methodological and evaluation choices in this study were made to maximize robustness, clinical relevance, and generalizability. The ensemble of ResNet18, DenseNet201, Xception, and InceptionV3 was selected because each architecture captures distinct retinal patterns, and their feature-level fusion with a Feedforward Neural Network enables stronger discriminative power than individual models or simple voting schemes. Preprocessing steps such as CLAHE, bilateral filtering, and Laplacian sharpening were applied to enhance image quality and preserve subtle pathological features, while augmentation mitigated class imbalance and improved generalization. The use of both a public dataset and a Bangladeshi clinical dataset ensured diversity and real-world applicability, and combining them further improved model stability. Finally, evaluation metrics were chosen beyond accuracy, including precision, recall, F1-score, specificity, and AUC, to provide a clinically meaningful assessment where false negatives are particularly critical.

4. Result analysis

4.1. Environment setup

All experiments were conducted using the PyTorch deep learning framework (version 1.13) with Python 3.9 in a Jupyter Notebook environment. The workstation used for training and evaluation was equipped with an AMD Ryzen 7–3800 CPU, 64 GB RAM, and an AMD Radeon RX 580 GPU with 8 GB VRAM, which provided sufficient computational capacity for handling high-resolution retinal images and training multiple deep learning models.

Torchvision and OpenCV were used for image preprocessing, augmentation, and loading pre-trained CNN architectures. NumPy supported numerical operations, while Matplotlib was used to visualize learning curves, ROC plots, and confusion matrices. This setup ensured a stable and efficient environment that enabled smooth model training, reliable evaluation, and reproducible results across the experiments.

4.2. Evaluation matrix

Several performance measures, which are accuracy, precision, F1 Score, recall and Specificity are used to measure the robustness of the proposed approach. This research uses the following performance evaluation criteria so as to determine whether the proposed investigation would be valuable. In that case, there may be a display of the confusion matrix that was used to evaluate the method's efficiency in relation to the next several indicators. The most applied, of course, depends on the confusion matrix, which is derived during testing for the identification task. Similar to predictability, Specificity is the proportion of all negative data that was correctly predicted and is expressed as a percentage [41]. All the performance evaluations were derived by using Eq. (23)–Eq. (27).

(i) Accuracy: The percentage of the overall accurate number based on the positive and negative classifications is known as accuracy.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{23}$$

(ii) Precision: The percentage of True Positives compared to all positive samples is known as precision.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{24}$$

(iii) F1-Score: The F1-score is an evaluation metric that brings together the precision and recall matrices into a single score by calculating the harmonic mean of both of these metrics.

$$F1\text{-}score = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{25}$$

(iv) Recall: Another term used for recall is sensitivity. The ratio of true positives to all positive samples is known as recall.

$$\text{Recall/Sensitivity} = \frac{TP}{TP + FN} \tag{26}$$

(v) Specificity: In the context of medical diagnosis, specificity refers to a test's capability to reliably identify individuals who do not have the disease (TN). Conversely, it is the percentage of correctly identified negatives.

$$\text{Specificity} = \frac{TN}{TN + FP} \tag{27}$$

(vi) AUC Score: One important performance indicator for classification tasks, particularly binary classification, is the AUC (Area Under the Curve) score. Plotting the True Positive Rate (TPR) against the False Positive Rate (FPR) at different thresholds, the Receiver Operating Characteristic (ROC) curve shows the area under the curve. A model's ability to differentiate between positive and negative classes is shown by its AUC.

4.3. Result evaluation

A detailed evaluation of the proposed ensemble-based model is conducted using three distinct datasets. Performance is measured using accuracy, precision, recall, F1-score, specificity, and AUC to assess the effectiveness of both individual CNNs and ensemble models. The analysis highlights how model integration and dataset diversity enhance diabetic retinopathy detection. Confusion matrices and ROC curves further illustrate classification accuracy and diagnostic reliability. To reduce the effect of overfitting and random variation, each experiment was repeated five times, and the mean performance with standard deviation (±) was reported for the ensemble models to provide a more reliable and stable evaluation. This approach provides greater clarity on the model's stability and robustness across multiple trials.

Standard deviation calculation

To evaluate the stability of the model across multiple runs, we compute the standard deviation using the following formula:

$$s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n - 1}} \tag{28}$$

where x_i denotes the accuracy from each run, $\bar{x}$ is the mean accuracy, and n is the number of runs.

Table 3
Performance Summary for the Dataset 1.

Used Models	Accuracy	Precision	Recall	**f1-Score**	Specificity	AUC Score
Resnet18	94.81	96.97	96.97	96.97	96.96	98.36
Densenet201	96.54	96.55	96.54	96.54	96.52	98.59
InceptionV3	97.40	98.27	98.27	98.27	98.27	98.82
Xception	97.40	97.84	97.84	97.84	97.83	99.51
ReXDen	98.10 ± 0.24	98.30	98.31	98.30	98.23	99.40
ReXInDen	98.27 ± 0.43	98.25	99.12	98.68	98.31	99.50

ReXDen = Ensemble of Resnet18, Xception and Densenet201.
ReXInDen= Ensemble of Resnet18, InceptionV3, Xception and Densenet201.

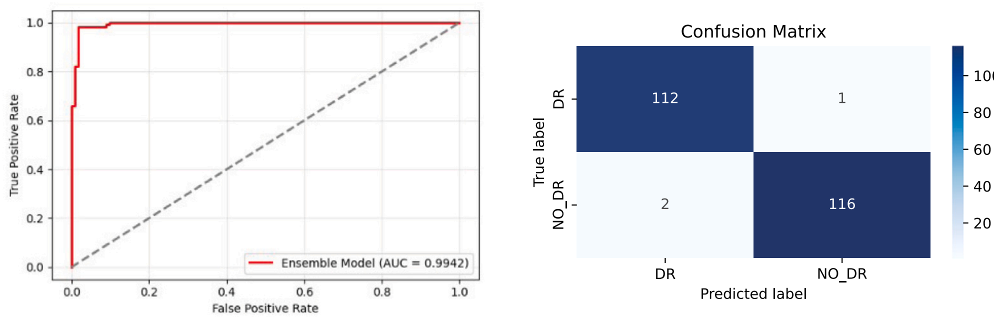

Fig. 10. Confusion matrix and ROC curve for the ReXInDen model on Dataset 1.

4.3.1. Evaluation results for dataset 1

Table 3 presents a comparative summary of the classification performance across multiple individual CNN models (ResNet18, Xception, InceptionV3, and DenseNet201) alongside two proposed ensemble approaches (ReXDen and ReXInDen) on Dataset 1. Among the individual models, InceptionV3 and Xception consistently outperformed ResNet18 and DenseNet201, achieving higher precision, recall, and AUC scores. InceptionV3 and Xception got the highest accuracy of 97.40% as individual models. However, the ensemble methods demonstrated clear improvements, highlighting the effectiveness of combining complementary architectures. Specifically, the ReXDen model achieved an accuracy of 98.10% ± 0.24 with an AUC of 99.40%, while ReXInDen achieved the best overall performance with an accuracy of 98.27% ± 0.43, recall of 99.12%, and the highest AUC score of 99.50%. These results suggest that ensemble integration not only reduces misclassifications but also enhances generalization, surpassing the performance of the best individual models.

Fig. 10 provides further insights through the confusion matrix and ROC curve of the proposed ReXInDen model. The confusion matrix indicates that out of all test samples, only three instances were misclassified (2 false positive and 1 false negative), while 228 cases were correctly classified (112 true positives and 116 true negatives). This demonstrates the model's strong diagnostic reliability in distinguishing DR from non-DR cases with minimal error. The ROC curve, with an impressive AUC score of 0.995, highlights the model's excellent discriminative ability across different classification thresholds. A near-perfect AUC value implies that the ensemble model is highly effective at maximizing sensitivity while minimizing false alarms, making it suitable for high-stakes medical screening tasks where both accurate detection and avoidance of unnecessary misdiagnosis are critical.

4.3.2. Evaluation results for dataset 2

Table 4 summarizes the classification outcomes for Dataset 2, highlighting both the strengths and weaknesses of individual CNNs compared to ensemble methods. Among the standalone models, ResNet18 and Xception maintained strong generalization with accuracies above 97.8%, while DenseNet201 also performed competitively with balanced precision and recall. In contrast, InceptionV3 struggled on this dataset, producing only 89.56% accuracy and a specificity of 93.69%, indicating that this architecture may not adapt as well to the variations in Dataset 2. The ensemble methods again surpassed the individual networks by integrating complementary decision boundaries. ReXDen reached an accuracy of 97.86% ± 0.28, while the more comprehensive ReXInDen achieved the best results, with 98.74% ± 0.12 accuracy and the highest AUC of 99.95%. These findings confirm that ensemble integration provides more robust and stable results by leveraging complementary strengths of different architectures.

The diagnostic capability of the proposed ReXInDen model is further demonstrated in Fig. 11. The confusion matrix shows that the model correctly identified 114 diabetic retinopathy cases and 265 non-DR cases, with only 2 instances each of false positives and false negatives. This very low error count illustrates the balance achieved between sensitivity and specificity. Importantly, the ROC curve highlights the near-perfect classification behavior of the model, achieving an AUC of 99.96%. Such a high score implies that across a wide range of thresholds, the ensemble maintains excellent separation between positive and negative classes.

4.3.3. Evaluation results for dataset 3

Table 5 provides the performance comparison on Dataset 3, which combines both local and publicly available samples to form a more diverse and challenging evaluation set. The individual CNNs achieved strong results, with DenseNet201, ResNet18, and Xception reaching accuracies above 98% and AUC scores above 99.6%. However, when tested on this more heterogeneous dataset, the superiority of ensemble learning became particularly evident. The ReXDen model achieved the highest accuracy of 99.05% ± 0.07, recall of 99.12%, and an AUC of 99.40%, clearly surpassing all single architectures. In contrast, the ReXInDen ensemble, though also highly effective, produced a slightly lower accuracy of 98.42% ± 0.09.

Fig. 12 presents the confusion matrix and ROC curve of the ReXDen model, which got the highest accuracy. Out of 610 test cases, the model correctly classified 225 DR cases and 380 non-DR cases, with only 2 false negatives and 3 false positives observed. This extremely low misclassification rate translates into high recall and specificity, minimizing both missed detections of DR and unnecessary false alarms. The ROC curve further illustrates the diagnostic strength of the model, with an AUC of 0.9941, confirming near-perfect separability between the two classes. The consistent alignment of high precision, recall, and AUC underscores that this ensemble is not just accurate but also reliable

Table 4
Performance Summary for the Dataset 2.

Used Models	Accuracy	Precision	Recall	f1-Score	Specificity	AUC Score
InceptionV3	89.56	94.26	94.18	94.21	93.69	97.92
Densenet201	97.65	96.62	96.61	96.61	96.10	98.39
Resnet18	97.91	97.91	97.91	97.91	97.28	99.62
Xception	98.17	98.17	98.17	98.17	98.47	99.82
ReXDen	97.86 ± 0.28	97.66	97.65	97.66	97.66	98.33
ReXInDen	98.74 ± 0.12	98.28	98.28	98.28	99.25	99.96

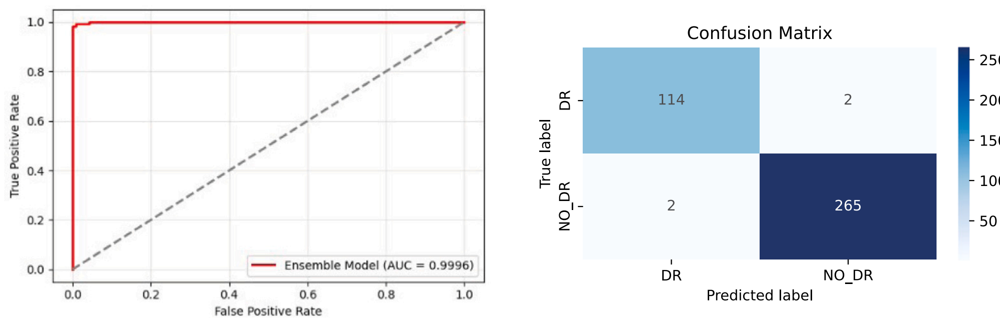

Fig. 11. Confusion matrix and ROC curve for the ReXInDen model on Dataset 2.

Table 5
Performance summary for the dataset 3.

Used Models	Accuracy	Precision	Recall	f1-Score	Specificity	AUC Score
InceptionV3	94.52	94.71	94.52	94.55	94.82	98.62
Densenet201	98.17	98.21	98.17	98.18	98.36	99.67
Resnet18	98.52	98.52	98.52	98.52	98.38	98.97
Xception	98.95	98.17	98.17	98.17	98.47	99.82
ReXDen	99.05 ± 0.07	98.68	99.12	98.90	99.22	99.40
ReXInDen	98.42 ± 0.09	98.34	98.36	98.44	98.34	98.43

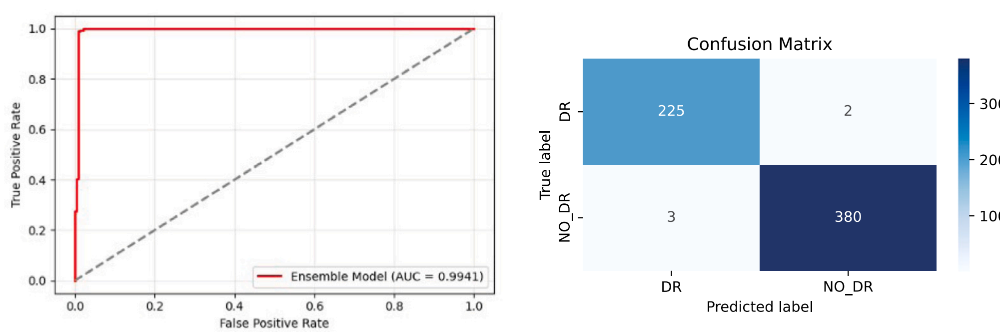

Fig. 12. Confusion matrix and ROC curve for the ReXDen model on Dataset 3.

across decision thresholds. As the best-performing approach across all experiments, ReXDen represents the robust and best solution for automated diabetic retinopathy screening in diverse clinical scenarios.

Across the three datasets, the performance of the two proposed ensemble frameworks showed a consistent but contrasting trend. The ReXInDen ensemble achieved the highest accuracy on Dataset 1 and Dataset 2, slightly exceeding ReXDen, which underscores the advantage of its expanded architecture. However, on the combined Dataset 3, ReXDen surpassed ReXInDen, achieving the best overall performance with 99.05% accuracy, along with high precision, recall, and AUC scores. This indicates that ensemble composition has a dataset-dependent influence on performance outcomes.

Fig. 13 highlights the superior performance of the proposed ensemble models across different datasets. The ReXInDen model attained 98.27% ± 0.43 accuracy on Dataset 1 and improved to 98.74% ± 0.12 on Dataset 2, showing not only higher accuracy but also reduced variability. Most notably, the ReXDen model achieved the best performance with 99.05% ± 0.07 accuracy on Dataset 3, establishing a new benchmark in both accuracy and stability. The narrow error margins further confirm the robustness and reliability of the proposed approach. These results emphasize the contribution of the ensemble design in advancing the detection of diabetic retinopathy with near-perfect precision.

To provide a concise overview of model performance across all datasets, Table 6 summarizes the best individual and ensemble results obtained from Table 3, Table 4, and Table 5.

The results in Table 6 clearly demonstrate that while strong accuracies were achieved by individual CNN models such as Xception and InceptionV3, the proposed ensemble frameworks consistently outperformed them in Dataset 1 and Dataset 2, with accuracy gains of +0.87% and +0.57%, respectively. On the combined dataset, the ReXDen ensemble achieved a peak accuracy of 99.05%, slightly surpassing the best individual model and exhibiting remarkable stability with minimal variation. These findings confirm that the ensemble

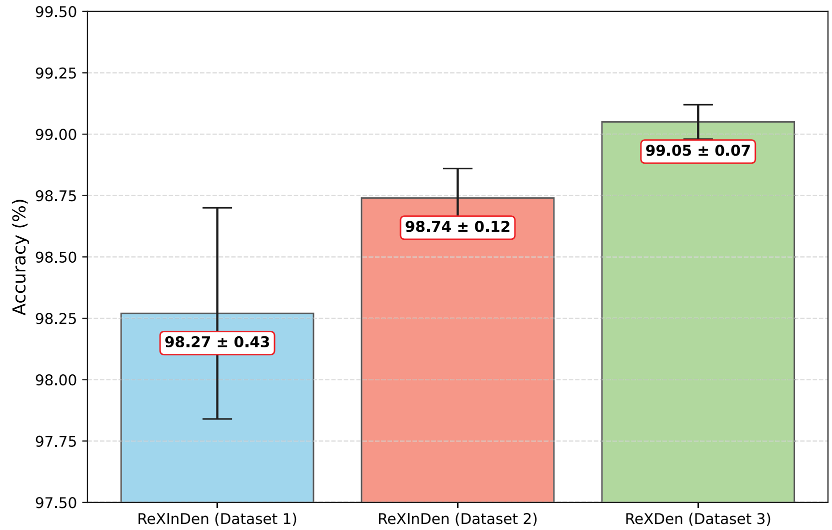

Fig. 13. Evaluation of Proposed Ensemble Architectures: Mean Accuracy with Standard Deviation.

Table 6
Summary of best individual vs ensemble model performance across three datasets.

Metric	KDRD	BD-DRiD	Combined
Best Individual Model Accuracy	97.40% (Xception/InceptionV3)	98.17% (Xception)	98.95% (Xception)
Best Ensemble Model Accuracy	98.27% (ReXInDen)	98.74% (ReXInDen)	99.05% (ReXDen)
Performance Gap (Ensemble - Individual)	+0.87%	+0.57%	+0.10%

approach effectively bridges the limitations of individual models, enhances robustness, and ensures strong generalization across diverse datasets.

4.4. Comparative result analysis

Table 7 indicates the accuracy for DR detection with models across different studies and compares it to the ensemble methods of the present study. However, Wong et al. and Sambit et al. utilized the ensemble method to get 96% and 96.98% accuracy on the APTOS dataset, whereas the ReXInDen model proposed by this study, which included an InceptionV3 along with DenseNet201, ResNet18, and Xception, got an accuracy of 98.27% on the Kaggle database. Pavate et al. used a MobileNet-based architecture model and achieved 95% accuracy on the Kaggle public dataset. Dohare et al. used an ensemble learning and got 98.5% outcome on accuracy.

In addition, the ensemble model, ReXInDen, showed a slightly improved accuracy of 98.74% on a set of Bangladeshi data alone which establishes the reality that ensemble approaches are quite effective when demographically specific data is used. Further, in this study, the ReXDen model achieved an accuracy of 99.05% on Dataset 3, which surpasses the maximum accuracy of 98.36% reported by Intifa Aman et al. In summary, this study confirms that current ensemble models show higher accuracy than most of the existing interventions and empirically proves the assurance of ensemble models for DR detection.

4.5. Discussion

The findings of this study confirm the effectiveness of ensemble-based CNN frameworks for diabetic retinopathy detection while also revealing nuanced insights into when such models achieve the greatest benefit. The ReXInDen model demonstrated superior performance on Dataset 1 and Dataset 2, with accuracies of 98.27% and 98.74%, respectively. This suggests that on relatively smaller or more homogeneous datasets, the inclusion of diverse backbones—particularly InceptionV3 with its multi-scale feature extraction—helps capture subtle lesion characteristics and improves generalization. In contrast, ReXDen slightly lagged in these datasets, indicating that the exclusion of InceptionV3 limited its ability to exploit fine-grained details under restricted data conditions.

Interestingly, this trend reversed on the more diverse and large-scale Dataset 3, where ReXDen achieved the highest accuracy of 99.05%. This outcome can be explained by the complementary strengths of ResNet18, Xception, and DenseNet201, which provided a balanced representation of both micro-level and macro-level retinal patterns without the redundancy introduced by InceptionV3. In larger datasets, where variation is naturally higher, excessive feature overlap from multiple backbones may hinder generalization, whereas a learner ensemble like ReXDen can avoid overfitting while still capturing the critical discriminative cues. This contrast underscores an important contribution of the present work: ensembles are not universally optimal, and their effectiveness depends on dataset size, complexity, and heterogeneity.

This trend may be attributed to the diminishing returns of adding more model complexity once sufficient data diversity and volume are present. In ensemble learning, increasing architectural diversity can improve performance on smaller or heterogeneous datasets by capturing varied feature representations, but it can also introduce redundant features and higher training complexity on large balanced datasets without corresponding gains in generalization. Similar phenomena have been reported in previous studies where more compact ensembles perform comparably or better than larger ensembles when data is abundant and well-distributed, as shown in work by Polikar [52] and Zhou [53], which discuss trade-offs between ensemble size, diversity, and performance in classification tasks. Therefore, the observed performance differences between ReXDen and ReXInDen are consistent with established ensemble learning behavior and suggest that dataset characteristics (size and balance) influence the optimal architecture choice.

While these high performances are encouraging, the exceptionally high accuracy of 99.05% requires critical reflection. Such results may indicate the possibility of overfitting or an overly favorable test set. Although rigorous pre-processing, augmentation, enhancement techniques, and multiple runs were employed to improve generalization, the clinical utility of the models can only be validated through external

Table 7
The comparison analysis of DR Detection.

Author	Model	Dataset	Accuracy
S. Pavithra et al. [1]	Ensemble Model (3 DL Models)	Kaggle	98.47%
Intifa et al. [15]	Hybrid Model (DenseNet121)	APTOS	98.36%
S.S. Mondal et al. [23]	Ensemble Deep Learning	APTOSDIABETDB1	96.98%
Wong. et al. [42]	ECOC Ensemble	APTOS	96%
Sonia et al. [43]	Ensemble	Private	97%
A. Ghulam et al. [44]	IR-CNN, Inception V3, ResNet 50	Kaggle	96.85%
Farag. et al. [45]	Automatic deep-learning approach	APTOS	97%
Dohare et al. [46]	Hybrid(GAN and BiGRU)	Kaggle	98.5%
A. S. Sabeena et al. [47]	MoNetPoly-NetXception	APTOS 2019	98.9%
T. Palaniswamy et al. [48]	IoTDL-DRD	MESSIDOR	98.14%
R.B. Dixit et al. [49]	EfficientNetB3	APTOS 2019	88.44%
K. Aurangzeb et al. [50]	Drive Chase-DB Stare	ColonSegNet	96.6%, 97.1%, 97.2%
S. Sundar et al. [51]	Neural Networks	Kaggle	90.57%
Proposed Model	ReXInDen	Dataset 1	**98.27%**
	ReXInDen	Dataset 2	**98.74%**
	ReXDen	**Dataset 3**	**99.05%**

testing on unseen datasets collected from independent populations. Without such validation, the risk remains that the reported accuracy may not fully translate into real-world screening environments where image quality and patient variability are less controlled.

Overall, this study demonstrates that ensemble learning can significantly improve diabetic retinopathy detection under certain data conditions, while also revealing that its effectiveness is not universal but dependent on dataset size and heterogeneity. By comparing ReXInDen and ReXDen, we show that more complex ensembles may excel on smaller, homogeneous datasets, whereas leaner ensembles can offer superior stability and generalization on larger, more diverse datasets. While the exceptionally high performance highlights the potential of our approach, its clinical utility must ultimately be confirmed through large-scale validation on external datasets. Future research will therefore focus on extending the dataset diversity, incorporating statistical validation methods, and exploring adaptive ensemble strategies to further enhance robustness and reliability.

5. Conclusion

Medical professionals can support the detection and treatment of diabetic retinopathy more effectively through the use of ensemble models, which contribute to the development of reliable diagnostic systems and ultimately improve patient outcomes. In this study, two ensemble frameworks, ReXDen and ReXInDen, were proposed to integrate the complementary strengths of multiple CNN architectures. Across three datasets, both ensembles consistently outperformed individual models, demonstrating the value of feature-level fusion in enhancing diagnostic accuracy and robustness. Among them, the ReXDen model achieved the highest performance on the combined dataset, reaching 99.05% accuracy, while the ReXInDen model showed strong results on smaller datasets with improved generalization. These findings confirm that carefully designed ensembles can advance medical image analysis by providing reliable tools for early diabetic retinopathy detection and screening.

Limitations and future study

Despite the encouraging results of our investigation, it is crucial to acknowledge certain limitations that may have impacted the outcomes. These consist of sample size limitations, potential dataset biases, and computational resource constraints. Future research will focus on addressing these limitations through the use of larger and more varied datasets, domain-specific expertise, and innovative DL architectures with post-processing procedures. To further improve classification performance, investigate state-of-the-art deep learning architectures such as transformer-based models or graph neural networks. Validation of a range of datasets and scenarios will be necessary to ensure that the proposed method is dependable and widely applicable. In addition, formal statistical validation will be incorporated through significance testing and resampling techniques to ensure that performance differences are scientifically robust. By tackling these issues, we intend to enhance the dependability and usefulness of our model, promoting medical image analysis and providing practitioners and healthcare institutions with useful information.

CRediT authorship contribution statement

Md Saykot Khandakar: The idea of this study, and research design, Data pre-processing and coding, Draft of the manuscript was prepared, Originally done. **Md Samsuddoha:** The idea of this study, and research design, Draft of the manuscript was prepared, Writing was reviewed, Supervision. **Sohely Jahan:** Result analysis and interpretation have been conducted, Writing was reviewed. **Rahat Hossain Faisal:** Result analysis and interpretation have been conducted, Writing was reviewed.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, we used Grammarly and ChatGPT in order to improve the quality of writing, make it more readable, and check grammar. After using these tools, we reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Declaration of competing interest

In order to ensure transparency and integrity in the research process, the authors state that there are no conflicts of interest.

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Finally,results were reviewed and approved by all authors and at the same time, all of them approved the final version of the manuscript.

Data availability

In the study, two unique datasets were used, one is accessible publicly at the following link: KDRD and another is collected manually from the Local Hospitals in Bangladesh: BD-DRiD. Respecting moral principles, confidentiality, and privacy are given top priority in all aspects of data gathering and use. The data and source code associated with this study are publicly available at: https://github.com/saykot55/Ensemble-based-Diabetics-Retinopathy-detection/.

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